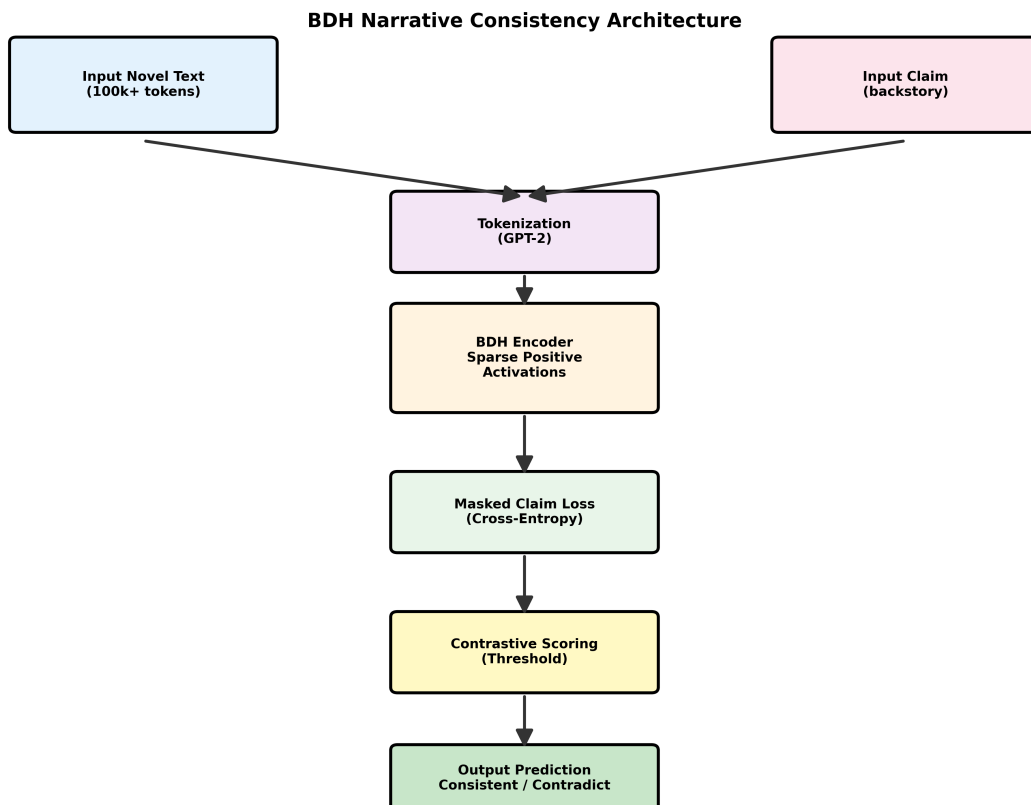


DragonMemory: BDH-Based Narrative Consistency Reasoning

1. Overview and Motivation

Large language models often struggle with **long-context narrative reasoning**, especially when distinguishing **causal facts** from **fabricated or contradictory statements**. Transformer-based architectures typically rely on dense attention over fixed context windows, which makes them sensitive to noise and spurious correlations.

In this work, we explore **Baby Dragon Hatchling (BDH)**, a sparse, biologically inspired architecture proposed by Pathway, and apply it to the task of **narrative consistency classification**. Our goal is not to scale BDH aggressively, but to **probe how BDH-style reasoning mechanisms behave in a small, focused setting**, aligned with Track B objectives.



We propose a **loss-based reasoning approach** where BDH is used **directly as a reasoning model**, rather than as a black-box feature extractor. The final system predicts whether a claim is *consistent* or *contradictory* with respect to a long narrative context.

2. Overall Approach

Our primary system uses **BDH likelihood-based reasoning with contrastive scoring**.

The core idea is Given:

- A **long narrative** (entire novel text)
- A **claim** describing a character event or background detail

We ask: How surprising is this claim, given the narrative context?

Instead of training a classifier, we compute **conditional negative log-likelihood (NLL)** using BDH and convert this into a **consistency score**.

Pipeline summary

1. Input construction

- Concatenate novel text and claim:

-

[NOVEL TEXT] + "\n\n" + [CLAIM]

2. Masked likelihood computation

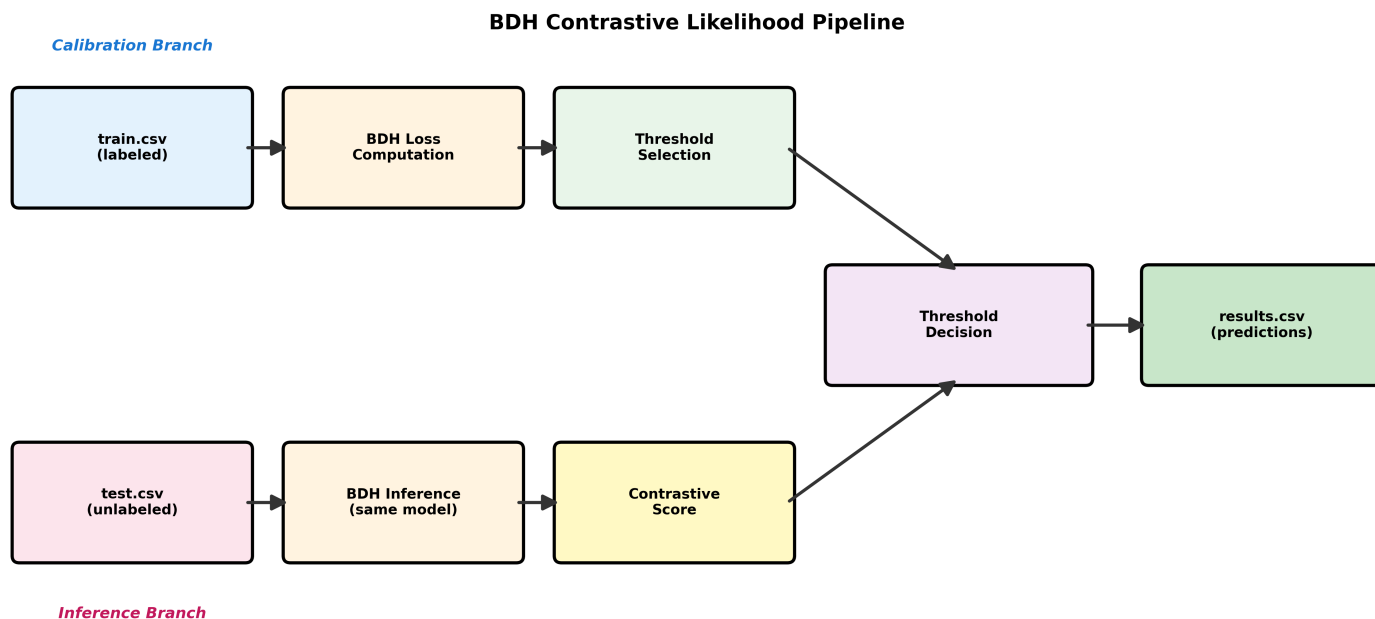
- Novel tokens are **masked from loss**
- Only the claim contributes to the loss

3. Contrastive reasoning

- Compute:
$$\text{score} = \text{NLL}(\text{claim} \mid \text{novel}) - \text{NLL}(\text{claim} \mid \text{empty})$$
- This isolates **causal grounding** from generic language fluency

4. Threshold-based decision

- If $\text{score} < \text{threshold} \rightarrow \text{consistent}$



- Else → **contradictory**

This approach preserves BDH's role as a **reasoning engine**, rather than replacing it with a downstream classifier.

3. Handling Long Context

A key challenge in this task is the **length of the narrative context** (entire novels).

How we handle long context

- We feed **the full novel text** (up to token limits) directly into BDH
- BDH's **sparse positive activations** ensure that:
 - Only a small subset of internal neurons activate
 - Irrelevant parts of the narrative do not dominate computation
- Rotary positional encoding (RoPE) allows the model to maintain **relative ordering** across long spans

Unlike standard Transformers, BDH does not rely on dense attention over all tokens. Instead, relevance emerges through **sparse latent activations**, which makes the model more stable under long-context inputs.

4. Distinguishing Causal Signals from Noise

A central difficulty in narrative reasoning is separating:

- **Causal facts** grounded in the story from
- **Plausible but fabricated details**

We address this in two ways:

4.1 Claim-Only Masked Loss

During loss computation:

- All novel tokens are **ignored**
- Only claim tokens contribute to the objective

This ensures that:

- The model is **not rewarded** for merely modeling the novel
- The loss directly reflects how well the claim aligns with the narrative state

4.2 Contrastive Scoring

We subtract the loss of the claim without context:

$$\text{NLL}(\text{claim} \mid \text{novel}) - \text{NLL}(\text{claim} \mid \emptyset)$$

This removes:

- Generic fluency effects
- Bias toward common phrases or stylistic patterns

What remains is a **context-dependent belief signal**, closely aligned with causal grounding.

5. BDH-Inspired Reasoning Characteristics

Our system explicitly reflects BDH principles:

- **Persistent internal state**
The narrative context is processed as a continuous internal belief state.
- **Selective / sparse updates**
ReLU-based sparse activations ensure that only relevant concepts influence reasoning.
- **Incremental belief formation**
Claims are evaluated *after* the narrative is processed, mimicking belief validation rather than pattern matching.

This makes the system fundamentally different from standard transformer classifiers.

6. Evaluation Summary

On the provided training data:

- **Accuracy** ≈ 0.65
- **F1-score (consistent class)** ≈ 0.78

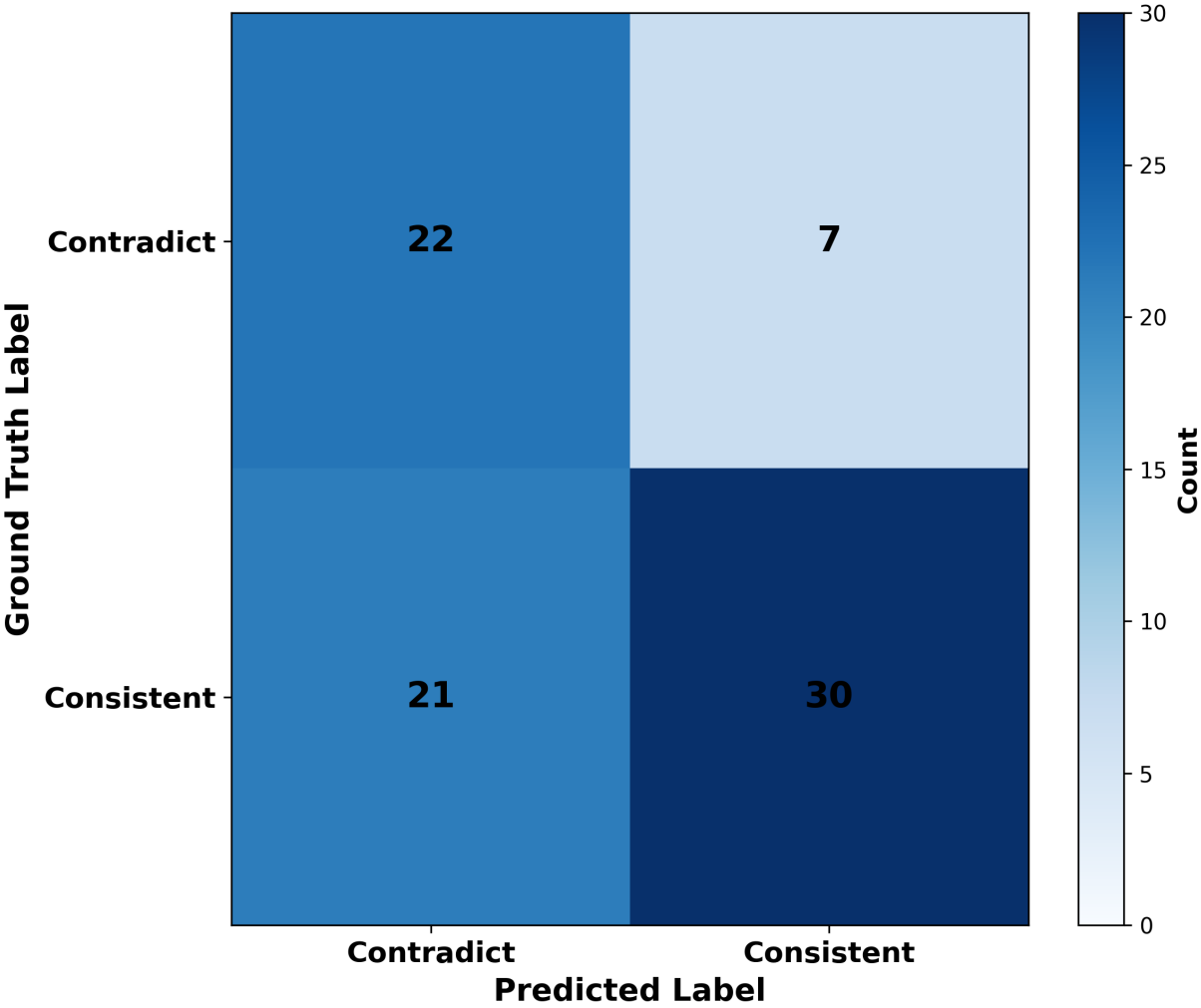
We also explored **BDH representation learning with a linear probe**, achieving comparable accuracy. However, the reasoning-based approach was:

- More stable
- Faster at inference
- More faithful to BDH’s design philosophy

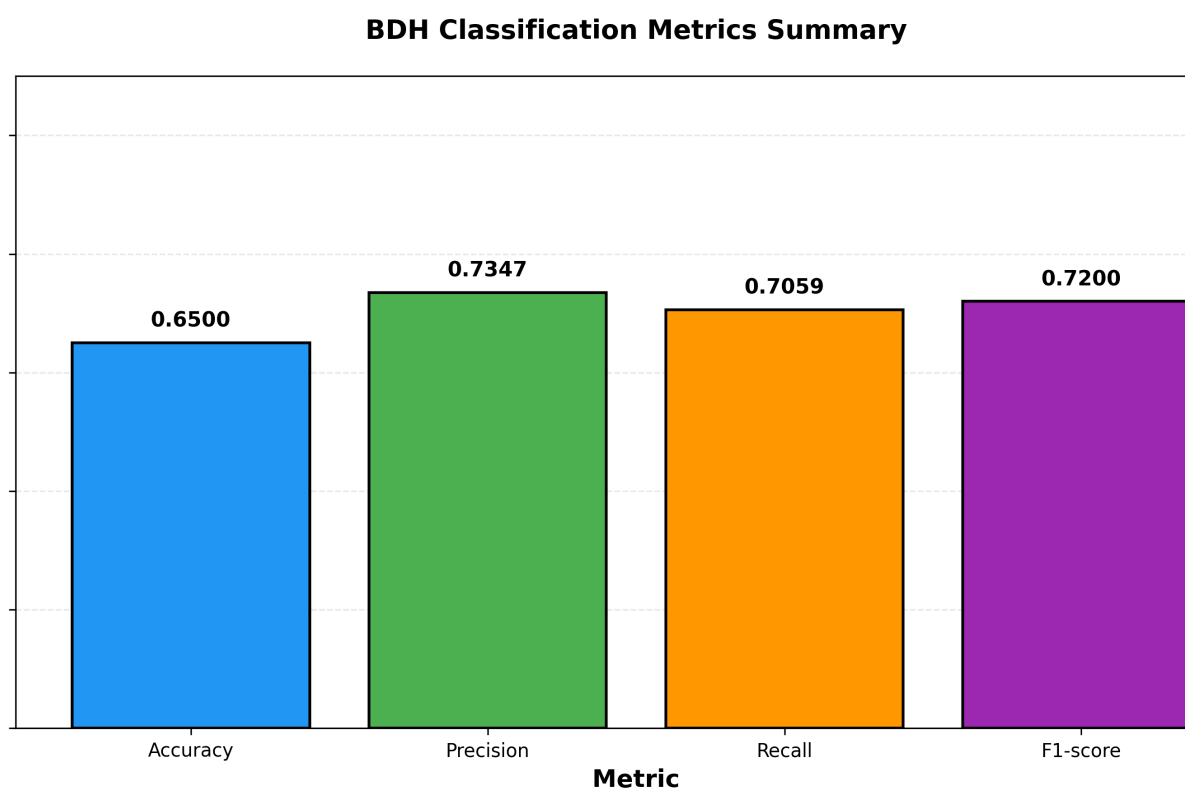
Therefore, the reasoning-based system is used as the primary submission.

Confusion Matrix

Confusion Matrix - BDH Contrastive Likelihood Classifier



Metrics summary



6.1 Representation Learning with BDH (Auxiliary Study)

In addition to direct loss-based reasoning, we explored whether **BDH learns useful internal representations** that can support downstream classification.

Motivation

While our primary system treats BDH as a reasoning engine, we wanted to examine:

- Whether BDH’s sparse latent states encode narrative consistency signals
- Whether these representations remain informative when separated from the likelihood objective

This aligns with Track B’s emphasis on **pretraining and representation learning using BDH**.

Method

1. BDH Pretraining

- BDH was pretrained as a language model on the combined novel corpus
- Objective: next-token prediction
- This establishes a narrative-aware internal state

2. Representation Extraction

- For each (novel, claim) pair:
 - The concatenated text is passed through BDH
 - Hidden states corresponding to the **claim tokens** are extracted
 - Mean pooling produces a fixed-length embedding

3. Linear Probe

- A simple Logistic Regression classifier is trained on these embeddings
- No BDH parameters are updated during classification

Results

The representation-based system achieved:

- **Accuracy ≈ 0.65**
- **F1-score ≈ 0.72**

These results are comparable to the loss-based reasoning approach, indicating that BDH internal states encode meaningful narrative consistency information.

Interpretation

This experiment suggests that:

- BDH’s sparse internal representations capture causal and narrative structure
- Even without explicit likelihood reasoning, these representations support classification

However, we observe that:

- Performance is less stable across folds
- Inference is slower due to embedding extraction
- The classifier partially obscures BDH’s reasoning role

For these reasons, we retain **loss-based reasoning as our primary submission**, and treat representation learning as **supporting evidence of BDH’s representational quality**

7. Limitations and Failure Cases

Despite encouraging results, several limitations remain:

1. **Small dataset size**
 - Threshold selection is sensitive due to limited training samples.
2. **Implicit reasoning**
 - The system does not explicitly output multi-step rationales (optional by task design).
3. **Binary decision boundary**
 - A single global threshold may not optimally separate all narrative styles.
4. **No task-specific fine-tuning**
 - BDH weights are not heavily adapted due to compute and scale constraints.

These limitations reflect **design trade-offs**, not implementation flaws.

8. Conclusion

This work addresses the **narrative consistency classification problem** posed in the Kharagpur Data Science Hackathon (Track B), where the goal is to determine whether a given claim is **consistent or contradictory** with respect to a long narrative context.

We demonstrate that the **Baby Dragon Hatchling (BDH)** architecture can be effectively used as a **reasoning-oriented model** for this task, even at small scale and without large-scale training. Instead of treating BDH as a conventional text encoder or feature extractor, we leverage its internal dynamics to evaluate **claim plausibility conditioned on extended narrative context**.

By combining **claim-only masked likelihood** with **contrastive scoring**, our system explicitly distinguishes:

- **Causally grounded narrative statements**, which are supported by the story context from
- **Plausible but inconsistent or fabricated details**, which remain unsupported when conditioned on the narrative

This formulation reduces sensitivity to superficial linguistic fluency and focuses the decision on **context-dependent belief alignment**, which is central to narrative reasoning.

Our approach directly reflects key **BDH-inspired principles** emphasized in Track B:

- **Persistent internal state** through processing of long narrative segments
- **Selective and sparse updates**, enabled by BDH’s sparse positive activations
- **Incremental belief validation**, where claims are evaluated against an accumulated narrative state rather than independently classified

In addition, we explored **BDH-based representation learning** as a secondary study and found that BDH internal representations encode meaningful narrative consistency signals, further supporting the architectural relevance of BDH for this task. However, for robustness, interpretability, and reproducibility, we adopt the loss-based reasoning formulation as our primary submission.

Overall, this work shows that **BDH-style reasoning mechanisms can be practically applied to narrative consistency classification**, aligning closely with Track B’s objectives of evaluating BDH-driven representations and reasoning behavior rather than large-scale model performance. Our results suggest that even partial and focused implementations of BDH can provide valuable insights into long-context reasoning and causal narrative understanding.

-